

LEARNING INTENTION

I can analyse how changing physical conditions in a habitat affect the growth and survival of living things.

SUCCESS CRITERIA

- ☐ I can identify physical conditions in a habitat
- ☐ I can explain how changes affect living things
- ☐ I can predict impacts of human activity on habitats

KEY CONCEPTS

Physical conditions: temperature, light availability, rainfall, salinity, soil type. These affect how well organisms grow and survive. **Human activity** (e.g. land clearing, pollution, artificial

WORKED EXAMPLE

Rising water temperature on the Great Barrier Reef causes coral bleaching. Coral expels the algae it needs for food → coral turns white → may die if temperature stays high.

PART A — WARM UP

1. Name FOUR physical conditions that can affect plant growth.
2. What is a habitat? Give one Australian example.
3. How might a drought (long period without rain) affect animals in a grassland?
4. Name ONE way human activity can change a habitat's physical conditions.

PART B — CORE SKILLS

5. A forest is cleared for a new suburb. List THREE physical conditions that would change as a result.
6. Bright artificial lights near a beach affect sea turtle nesting. Explain which physical condition has changed and predict the impact on turtles.
7. How could increased salinity (salt levels) in a river affect the plants living there?
8. Show your working/reasoning for Question 5 below.

SCIENCE HELPER

Living things have a **tolerance range** for physical conditions — conditions that are too extreme (too hot, too cold, too salty, too dry) will prevent growth or cause death. Even small changes in physical conditions can have big impacts on ecosystems over time.

WORKING SPACE — NOTES AND EVIDENCE

PART C — CHALLENGE

9. Real-World Context: Climate change is causing temperatures in the Australian Alps to rise, reducing the amount of snow each year. The mountain pygmy possum relies on deep snow for insulation during winter hibernation. Using the terms physical conditions, habitat, survival and human activity, explain the threat this poses and one possible action humans could take.

10. Reasoning: A student grows two identical bean plants — one in full sunlight and one in a dark cupboard. After two weeks, the plant in the dark is pale, thin and weak. Explain why, naming the physical condition that changed and what effect this had on the plant's ability to survive.

REFLECTION

Today I learned...

I CAN CHECKLIST

- ☐ I can identify physical conditions in habitats
- ☐ I can explain how changes affect living things
- ☐ I can predict impacts of human activity

TEACHER FEEDBACK

Strength: _____

Next Step: _____

PARENT TIP

Grow two plants — one in sunlight, one in shade. Observe the differences over two weeks together.

TEACHER TIP

Use Australian case studies (Great Barrier Reef, alpine habitats) to connect science to real local conservation issues.

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: Temperature, light, rainfall/water, soil type (accept salinity, humidity)

Q2: Natural environment where organisms live; e.g. Great Barrier Reef, Daintree Rainforest, dry outback

Q3: Water sources dry up; food (plants) dies; animals may migrate or die of thirst/starvation

Q4: Land clearing, pollution, artificial lighting, irrigation (accept any one)

Q5: e.g. Reduced shade → higher temperature; reduced rainfall interception → drier soil; less canopy → more direct sunlight reaching ground

Q6: Light availability changed; turtles use darkness to navigate to nesting sites — artificial light disorients

them, reducing successful nesting

Q7: Most freshwater plants cannot tolerate high salinity — they may wilt, die or fail to reproduce

Q8: Working space provided

Q9: Rising temperatures (human-caused climate change) reduce snowfall → loss of insulation → possum cannot hibernate safely → threat to survival. Action: reduce carbon emissions, create artificial snow habitat, translocation programs

Q10: Light availability was reduced in the cupboard. Without light, the plant cannot photosynthesise → cannot produce food/energy → grows weak and pale (etiolation); without energy it cannot survive long-term